Experiment - 13

IMPLEMENT YOUR OWN PING PROGRAM.

Aim:

To implement a ping program using python raw sockets to check host reachability of a host and calculate the round trip time.

Code:

```
import socket, struct, time, os

def checksum(data):
    total = 0
    for i in range(0, len(data)-1, 2):
        total += (data[i] << 8) + data[i+1]
    if len(data) % 2:
        total += data[-1] << 8
    total = (total >> 16) + (total & 0xffff)
    total += total >> 16
    return (~total) & 0xffff

def ping(host):
    try:
        dest_addr = socket.gethostbyname(host)
        soch = socket.socket(socket.AF_INET,
                             socket.SOCK_RAW,
                             socket.IPPROTO_
                                 ICMP)
        packet_id = os.getpid() & 0xffff
        sequence = 1
        header = struct.pack(
            "!BBHHH",
            8,
            0,
            packet_id, sequence)
```

```
data = b"PING"
chksum = checksum(header + data)
header = struct.pack(
        "!BBHHH",
        8,
        0,
        checksum,
        packet_id,
        sequence)
packet = header + data
start = time.time()
sock.sendto(packet, dest_addr, 1))
sock.settimeout(2)
sock.recvfrom(1024)
end = time.time()
rtt = (end - start) * 1000
print("Reply from", dest_addr,
                "time=%.2fms", % rtt)
sock.close()
except socket.timeout:
        print("Request timed out")
except PermissionError:
        print("Permission denied: Run the
    program as administrator (root.")
host = input("Enter destination host:")
ping(host)
```

Sample INPUT/OUTPUT:

Enter destination host: 8.8.8.8

Reply from 8.8.8.8: time = 24.53 ms

Result:

Thus the ping program was successfully implemented in python using raw sockets.